

Nature Communications v2 — Expanded Methods, CRediT, Ethics, Reporting Summary

Paper 1 · Cross-ethnic 8-gene RAI-lineage panel · Cook S (국승호) & Yu H-W · compiled 2026-05-21
 Accompanies main bioRxiv preprint (paper1_biorxiv.html) and NComms submission package (ncomms_submission_package.html).

0. Figure trim plan — 23 main → 8 main + 15 supplementary (NComms 10-figure limit)

NComms Main (8)	Source figure (current draft)	Rationale
Fig 1. 3-tier architecture overview	Fig 1 + Fig 2 (merged)	Single overview frame: TIERA hierarchy + 2-layer architecture
Fig 2. Cross-cohort master forest (13 entries)	Fig 11b	Centerpiece — cross-ethnic + cross-modality consensus in one frame
Fig 3. Biology-stratified DM1 mega-comparison	Fig 5b + Fig 5c (merged)	Driver + histology + axis stratification in 6 cohorts
Fig 4. Fusion enrichment + clinical	Fig 7 (Table 4 as inset) + Fig 10 (PFI/OS)	76.8% / OR 7.41 + honest PFI Cox PH
Fig 5. Epigenetic mechanism cascade	Fig 18 + Fig 19 (merged)	HM450 per-gene + mechanism cascade
Fig 6. Single-cell thyrocyte-intrinsic + DepMap	Fig 20	Q7 immune-confound knockout + druggable hub
Fig 7. PRISM drug screen — MAPK-inhibitor sensitivity	Fig 21	Translation hook (NComms editor magnet)
Fig 8. Graphical abstract / integrated model	fig_graphical_abstract	Required for NComms broad readership

Supplementary figures (15 total)

- S1. RAI biology pathway pretty (Fig 3)
- S2. TIERA67 gene-set hierarchy (Fig 2 detail)
- S3. Cross-cohort panel scan TCGA + K2 + Lee + GSE286332 (Fig 4)
- S4. External GPL570 direction-consistency forest (Fig 6)
- S5. Zero-overlap NONOVERLAP₈ scatter grid (Fig 7)
- S6. External histology boxplots (Fig 8)
- S7. TCGA-only 16-panel scan (Fig 5)
- S8. Quantum kernel SVM honest disclosure (Fig 9)
- S9. DM1 robustness forest 12 contrasts (Fig 11)
- S10. Per-gene heatmap (Fig 12)
- S11. Knock-out n-1 sensitivity (Fig 13)
- S12. ROC + KM + Calibration + DCA quad (Fig 14-17)
- S13. Korean cohort applicability board (Supp 2)
- S14. LOO sensitivity combined (Supp 1)
- S15. Image-DM1 TSS-confound case study (Supp Note)

1. Expanded Methods (NComms unlimited-length)

1.1 Patient cohorts and ethics

This study uses exclusively publicly available, previously published datasets. No new patient samples were collected; all data were re-analysed under the original published data-use agreements. No institutional review board approval was required for the secondary analysis of de-identified public datasets. The following 10 distinct cohorts were used:

Cohort	Accession	n primary	Platform	Use in this study	Original publication / Data-use authority
TCGA-THCA	GDC thca_tcga_pan_can_atlas_2018	504	Illumina HiSeq RNA-seq + HM450 methylation array	Discovery DM1/DM2 partition, fusion SV, methylation, PFI/OS	TCGA Research Network Cell 2014
MSK-IMPACT thyroid	Landa 2016 supp / cBioPortal	117	Targeted-panel + clinical follow-up	OS replication in advanced disease	Landa et al. JCI 2016
Korean K2 / PRJEB11591	ENA PRJEB11591	260 RNA-seq	Illumina RNA-seq	Cross-ethnic Korean validation	Yoo et al. PLOS Genet 2016
Korean GSE286332	GEO GSE286332	18 (PTC 9 + PTC+HT 9)	RNA-seq FPKM	Ground-truth-labelled Korean	Korean PTC vs PTC+HT cohort 2024
Lee 2024 / GSE213647	GEO GSE213647	630 post-QC	RNA-seq	Independent Korean cohort replication	Lee et al. 2024 Korean (in press)
Mun 2025 proteogenomics	Mun 2025 supp	336	Mass-spectrometry proteomics	Cross-modality protein replication	Mun et al. Cell 2026 (in press)
Landa 2016 GSE76039	GEO GSE76039	37 (PDTC/ATC)	Affymetrix microarray	Western advanced-disease replication	Landa et al. JCI 2016
External GPL570 × 4	GSE29265, GSE33630, GSE53157, GSE65144	287 total	Affymetrix HG-U133 Plus 2.0	External direction-consistency	Tomas 2012/2014, Salvatore 2014, von Roemeling 2015
Lu 2023 single-cell	GEO GSE193581	67,678 cells / 23 patients	10x Genomics 3' v3	Single-cell thyrocyte-intrinsic axis	Lu et al. Cancer Cell 2023
DepMap public	DepMap 23Q4 release	1,141 cell lines / 11 thyroid	CRISPR essentiality (Chronos) + PRISM drug-screen	Druggable target stratification	Broad Institute DepMap consortium

1.2 Panel definitions and provenance

The 8-gene RAI-lineage panel (RAI₈: SLC5A5/NIS, TPO, TG, TSHR, PAX8, NKX2-1, FOXE1, DIO1) was derived by RandomForest feature-importance ranking (sklearn 1.5, default 100 trees, gini criterion) on a 55-gene TIERA67-clean candidate pool (TIERA67 67-gene curated framework with Driver_anchor 12-gene category removed; documented in [project/notebooks_or_scripts/rerun_v2.py:121-200](#)) using TCGA-THCA RNA-seq z-scores against the d4p2 reference DM1/DM2 labels. The 3-gene TF₃ subset (FOXE1, NKX2-1, PAX8) corresponds to the developmental master TF

triad (Damante & Di Lauro series); the 8-gene zero-overlap NONOVERLAP₈ (SLC26A4, IYD, DUOX1, DUOX2, TFF3, HHEX, GLIS3, DIO2) was hand-curated from the TIERA67 candidate pool sharing zero overlap with RAI₈.

1.3 Within-cohort z-score panel scoring

For each cohort C and gene g in a panel P , the within-cohort z-score is $z_{C,g,i} = (x_{C,g,i} - \text{mean}_C(x_{C,g})) / \text{sd}_C(x_{C,g})$ where x is $\log_2(\text{TPM}+1)$ for RNA-seq cohorts and platform-normalised $\log_2$ intensity for microarrays. The panel score for sample i is the arithmetic mean across panel genes: $\text{panel_}z_i = \text{mean}_{g \in P}(z_{C,g,i})$. The within-cohort normalisation step removes ethnicity-correlated mean shifts and platform bias to first order; we explicitly verify the same axis emerges under cohort-pooled ComBat batch correction (Johnson et al., Biostatistics 2007) for sensitivity.

1.4 Within-cohort unsupervised k=2 KMeans clustering

For each cohort, sample-level panel z-scores (8-dimensional for the full 8-gene panel, or per-gene z values stacked) were clustered with sklearn KMeans ($n_clusters=2$, $n_init=10$, $random_state=42$). The cluster with the lower mean $\text{panel_}z$ was assigned as DM1 (silenced lineage) by biological convention. Cohen's d between clusters was computed as $|\text{mean}(\text{DM1}) - \text{mean}(\text{DM2})| / \text{pooled_SD}$ where pooled_SD uses the unbiased two-sample pooled formula. This procedure is fully unsupervised within each cohort and does not use any TCGA-derived label transfer.

1.5 Supervised classification — LogReg, RBF-SVM, quantum kernel SVM

Logistic regression: sklearn 1.5 LogisticRegression with L2 penalty and default regularisation strength $C=1.0$; stratified 5-fold cross-validation; reported AUC is mean across folds. **RBF-SVM:** sklearn SVC with $\text{kernel}="rbf"$, $\text{gamma}="scale"$, $C=1.0$; same stratified 5-fold CV protocol. **Quantum kernel SVM:** PennyLane 0.44.0 with default.qubit backend; Havlíček 2019 ZZ-feature-map (Nature 567:209) with $\text{reps}=2$ and one qubit per panel feature (3-8 qubits depending on panel size); $n=80$ stratified subsample due to circuit cost; quantum kernel matrix computed via inner-product of feature-mapped states.

1.6 Meta-analysis methods

DerSimonian-Laird random-effects model (REM): inverse-variance random-effects pooling of cohort-level AUC values across k cohorts with τ^2 estimation. **Han-Eskin RE2 (Han & Eskin, AJHG 2011):** threshold by Cochran's Q I^2 ($I^2 < 30\% \rightarrow \text{fixed effect}$; $I^2 \geq 30\% \rightarrow \text{REM}$). **Stouffer's weighted Z:** sum of $\sqrt{n}$ -weighted per-cohort Z-statistics, direction-consistent. **ComBat empirical Bayes:** Johnson 2007 method applied to TCGA + K2 expression matrices before pooled scoring; ComBat-Seq variant available for count-based future extensions. **LOCO sensitivity:** leave-one-cohort-out pooled AUC across the four labelled cohorts (TCGA, K2 $n=260$, K2 balanced $n=34$, GSE286332).

1.7 Robustness battery

Permutation null: 5,000 random shuffles of DM1/DM2 labels with panel fixed; empirical $p = (1 + \#shuffles \geq \text{observed AUC}) / 5,001$. **Panel-conditional null:** 10,000 label shuffles, panel fixed, tracking $\max |\text{Cohen's } d|$ under shuffled labels. **Thyroid-matched bio-prior null:** 1,000 random 8-gene draws from the top-quartile thyroid-expressed gene pool ($n=12,921$ genes, defined as TCGA-THCA mean TPM > 75th percentile), recomputing the within-cohort z-mean panel score and Cohen's d each time. **Knock-out n-1 floor:** drop each panel gene in turn and recompute the DM1 vs DM2 Cohen's d . **Bootstrap cluster stability:** 500 bootstrap iterations of the d4p2 sig_score distribution with KMeans $k=2$ re-clustering; per-sample stability = fraction of iterations the sample is assigned to its original cluster.

1.8 Clinical / survival analysis

Cox proportional hazards regression was performed using lifelines 0.30 (Davidson-Pilon Python package). The multivariable model adjusted for age (continuous, per year), sex (binary, male reference), AJCC pathologic stage (ordinal score 1-4 from Stage I through Stage IV), BRAF V600E status (binary), and RAS status (binary). Endpoints: progression-free interval (PFI, primary; events 11.6%) and overall survival (OS, secondary; events 3.4%). DM1 vs DM2 covariate added as the variable of interest. The Cox PH assumption (proportional hazards across covariates) was verified by Schoenfeld residual tests (Therneau-Grambsch); no significant violations were detected. **Pooled meta-analysis** of TCGA + MSK-IMPACT OS HR estimates used DerSimonian-Laird random-effects (statsmodels 0.14 meta_analysis module); between-cohort heterogeneity assessed via Cochran I^2 . **NRI / IDI** computed under 5-fold stratified-CV LogReg probabilities per Pencina 2008 continuous-NRI definition (event = DM1, non-event = DM2; equal weighting between cases and controls); 95% CI from 2,000 stratified bootstrap iterations of test-fold predictions. **Decision curve analysis** per Vickers & Elkin 2006 on TCGA 5-fold CV probabilities.

1.9 HM450 methylation analysis

TCGA-THCA HumanMethylation 450K array β -values were downloaded from GDC for n=503 primary tumours. Promoter probe-set summarisation used the mean β -value across probes within 1500 bp upstream of the canonical transcript start site (Ensembl GRCh38 annotation) per gene. Per-gene Cohen's d between DM1 (n=87) and DM2 (n=25) was computed using the unbiased two-sample pooled standard-deviation formula; significance was assessed by Wilcoxon rank-sum test with Benjamini-Hochberg FDR correction across the 8 panel genes. Within-DM1 fusion+/- stratification used cBioPortal SV calls.

1.10 cBioPortal SV / fusion analysis

Structural-variant calls were downloaded via the cBioPortal REST API for the `thca_tcga_pan_can_atlas_2018_structural_variants` molecular profile (n=542 SV-tested primary tumours). Fusion-positive tumours were defined as harboring at least one fusion involving RET, NTRK1, NTRK3, ALK, or BRAF as either the 5' or 3' partner. Missingness was assessed via chi-square MAR test against clinical features (p=0.56, MAR confirmed). Sensitivity analyses imputing missing-as-negative, missing-as-prevalence-matched, and complete-case all retained the DM1 vs DM2 OR in the range 7.18-9.07.

1.11 Single-cell analysis (Lu 2023 GSE193581)

Pre-processed single-cell RNA-seq data (15,330 thyrocytes/epithelial cells from 21 patient samples; 67,678 total including non-tumour) were downloaded from `project/results/v17_lu2023/GSE193581_cell_panel_score.tsv`. The 8-gene panel z-score (DM_score in the source TSV) was computed per cell using within-cohort z-mean of normalised log-counts. Per-patient mean DM_score in malignant cells was used for the forest plot in Fig 6B. Cohen's d between ATC vs PTC malignant cells (n=6,034 + 8,590 cells) was computed at the single-cell resolution.

1.12 DepMap CRISPR + PRISM analysis

DepMap 23Q4 release: 1,141 cell lines $\times$ CRISPR Chronos essentiality scores were stratified by DM1_like_score (computed via the 8-gene panel z-mean from CCLE expression data per `paper9_sl_first_pass/ccle_dm1_state.tsv`); DM1-high vs DM1-low groups defined by median split (n=372 vs n=394). Per-gene Δ effect (DM1-high – DM1-low) was computed with Welch t-test and Cohen's d effect size. PRISM drug screen log-fold-change values for 1,518 compounds tested across 470-480 cell lines were stratified by the same DM1-state grouping; FDR correction by Benjamini-

Hochberg. **Thyroid-lineage-specific subset** (n=11: CAL62, BHT101, BCPAP, TT2609C02, FTC133, 8305C, 8505C, MB1, ASH3, IHH4, HOTH4) absolute essentiality medians were extracted from [/data/thca/repo_results/paper11_pancancer/phase_D_depmap/lineage_dependency_summary.tsv](#).

1.13 Statistical reporting

All p-values are two-sided unless explicitly stated as one-sided. Effect sizes use Cohen's d with the unbiased two-sample pooled standard-deviation formula. 95% confidence intervals for Cohen's d use bootstrap resampling (2,000 iterations stratified by class). 95% CIs for Cox hazard ratios use the Wald approximation on log-HR. Multiple-testing correction uses Benjamini-Hochberg FDR within each analysis family (e.g., 8-gene HM450, 16-panel scan, 12-contrast effect-size matrix). Exact n per panel/test/comparison is reported in each figure caption.

1.14 Software versions and reproducibility

Library / tool	Version	Use
Python	3.12	Analysis platform
pandas	2.x	Data manipulation
numpy / scipy	1.x / 1.14	Numerical / statistical
scikit-learn	1.5	Clustering, classification, KMeans
lifelines	0.30	Cox PH survival analysis
statsmodels	0.14	Meta-analysis, statistical tests
matplotlib	3.x	Figure generation
anndata / scanpy	0.10 / 1.10	Single-cell analysis
PennyLane	0.44	Quantum kernel SVM
WeasyPrint	68.1	PDF rendering
kallisto	0.51.1	K2 RNA-seq quantification
GENCODE	v44	K2 reference annotation

2. Author contributions (CRediT taxonomy)

Conceptualization: S.C., Y.H.-W. — Y.H.-W. proposed the BRAF/RAS-negative dark-matter clinical positioning; S.C. designed the selection-layer framework, 3-tier panel architecture, and cross-ethnic robustness pre-specification.

Data curation: S.C. — TCGA-THCA, MSK-IMPACT, K2 PRJEB11591, GSE286332, GSE213647, GSE76039, GSE29265, GSE33630, GSE53157, GSE65144, GSE193581, Mun 2025, DepMap, PRISM acquisition and harmonization.

Formal analysis: S.C. — all statistical analyses, including discovery clustering, panel scanning, meta-analysis, robustness battery, Cox PH, NRI/IDI, single-cell, DepMap, PRISM. Y.H.-W. clinical interpretation review.

Funding acquisition: Y.H.-W. — institutional support [grant numbers TBD at submission].

Investigation: S.C. — all in silico investigations. Y.H.-W. domain expertise on thyroid clinical context.

Methodology: S.C. — designed the within-cohort z + unsupervised KMeans approach, the K2 calibration-drift diagnostic, the layered panel architecture, the Korean cohort transfer-label limitation framing, the multi-method meta-analysis battery, and the single-cell/DepMap/PRISM stratification protocol.

Project administration: S.C., Y.H.-W.

Resources: Y.H.-W. — clinical computing resources (Seoul National University Bundang Hospital). S.C. — public dataset infrastructure.

Software: S.C. — all analysis pipelines, panel scanning code, figure-build scripts; public release tag `paper1-ncomms-submission-2026-05-21` at github.com/kukshomr/THCA_data_analysis.

Supervision: Y.H.-W. — clinical translation supervision, cohort selection direction, reviewer-anticipation strategy.

Validation: S.C., Y.H.-W. — S.C. computational cross-checks (pre-submission verification status box catalog); Y.H.-W. clinical plausibility check.

Visualization: S.C. — all 23 main + 2 supplementary figures; graphical abstract.

Writing — original draft: S.C. — Results, Methods, Reviewer Q&A. Y.H.-W. — voice-protected sections (§ 1.1 Hook, § 3.1 Discussion opening, § 3.4 Limitations, Q9, Cover letter ¶ 1).

Writing — review & editing: S.C., Y.H.-W. — collaborative final review.

3. Inclusion & ethics statement

Cohort inclusion rationale. Korean cohorts (K2 PRJEB11591, GSE286332, Lee 2024/GSE213647) are explicitly included because thyroid cancer incidence is disproportionately concentrated in East Asia (South Korea has the world's highest reported incidence), and prior Western-cohort-only molecular taxonomies inadequately characterize the BRAF/RAS-negative compartment in this population. Cross-ethnic validation is the deciding analysis pre-specified in § 1.4 of the main text. We acknowledge that Japanese, Chinese, and Taiwanese RNA-seq cohorts of comparable depth were not available at the time of writing; a true Pan-Asian extension is reserved for the planned Paper 4 backlog.

Data-use compliance. All datasets used in this study are publicly available and were re-analysed under their original data-use agreements. TCGA-THCA and MSK-IMPACT (Landa 2016) data were accessed through GDC and cBioPortal with appropriate institutional research-use authorisation. ENA PRJEB11591 (K2) was accessed under its standard public re-use terms. GEO datasets (GSE286332, GSE213647, GSE76039, GSE29265, GSE33630, GSE53157, GSE65144, GSE193581) were accessed under GEO's public re-use terms. DepMap and PRISM data are released by Broad Institute under permissive CC BY 4.0. No new patient samples were collected for this study; no human-subject institutional review board approval was required for the secondary analysis of de-identified public datasets.

Authorship and contributorship. S.C. (국승호, Seungho Cook) is a graduate-level researcher; Y.H.-W. (유형원, Yu Hyeong-won) is a senior clinician-investigator at Seoul National University Bundang Hospital. Both authors are based at SNUBH and meet the four ICMJE authorship criteria. No paper was contracted out for ghostwriting or AI prose generation in voice-protected sections (§ 1.1 Hook, § 3.1 Discussion opening, § 3.4 Limitations, Q9, Cover letter ¶1); these sections require author-keyboard input per our internal voice-protection policy.

Conflict of interest. The authors declare no competing financial or non-financial interests related to this work.

Funding. Institutional support from Seoul National University Bundang Hospital. Specific grant numbers TBD at submission.

4. Nature Research Reporting Summary (draft answers)

4.1 Statistics

n described: Yes — exact n per panel/test/comparison reported in each figure caption and Table 1-6.
Statistical test details: Yes — see Methods § 1.13 + figure captions; all tests two-sided unless noted.
Adjustment for multiple comparisons: Yes — Benjamini-Hochberg FDR within analysis families.
Software used to compute statistics: Yes — pandas 2.x, scipy 1.14, statsmodels 0.14, lifelines 0.30, scikit-learn 1.5, PennyLane 0.44.
Are data well-described: Yes — sample sizes, demographics, cohort sources tabulated in Table 1 + Supp Table S1.

4.2 Software and code

Code availability: Public GitHub repository at github.com/kukshomr/THCA_data_analysis, release tag [paper1-ncomms-submission-2026-05-21](#), archived at Zenodo with DOI [placeholder, mint at submission].
Custom code: All analysis scripts in [project/notebooks_or_scripts/](#) and [scripts/boundary_audit/](#) are deposited as open-source MIT license.
Algorithms used: RandomForest (sklearn), KMeans (sklearn), LogisticRegression (sklearn), SVC RBF (sklearn), Quantum kernel ZZ-feature-map (PennyLane), DerSimonian-Laird REM, Han-Eskin RE2, ComBat empirical Bayes, Pencina continuous-NRI, Vickers-Elkin DCA.

4.3 Data

Accession codes/IDs: TCGA-THCA (GDC), MSK-IMPACT thyroid (Landa 2016 supp + cBioPortal), PRJEB11591 (ENA), GSE286332, GSE213647, GSE76039, GSE29265, GSE33630, GSE53157, GSE65144, GSE193581 (all GEO), Mun 2025 (Cell supp), DepMap 23Q4 (depmap.org).
Restricted-access data: None used.
Data availability statement provided: Yes — see § 3 above and Methods § 1.1.
Data deposition at Nature Research-affiliated archive: Korean K2 + Lee + GSE286332 unsupervised analysis outputs deposited at Zenodo alongside code.

4.4 Field-specific reporting

Life sciences: Selected. Reporting summary covers cell-line / patient-derived data quality.
Behavioural & social sciences: N/A.
Ecological / evolutionary / environmental: N/A.
Materials & reagents: Public datasets only; no custom reagents synthesized.
Antibodies: N/A.
Cell lines: DepMap public cell-line collection (1,141 lines × CRISPR + 470-480 lines × PRISM). All identities authenticated by Broad DepMap consortium (CCLE).
Sex / gender: Sample-level sex covariate included in multivariate Cox PH (Methods § 1.8); sex distribution per cohort tabulated in Supp Table S1.
Ethnicity: Cross-ethnic validation is the central methodological commitment; Korean (PRJEB11591, GSE286332, GSE213647) and Western (TCGA, MSK, GPL570 series) cohorts are explicitly stratified and reported separately.
Replication: 13 entries across 10 cohorts × 4 modalities provide replication at the cross-cohort and cross-modality level.
Randomization: Bootstrap iterations and KMeans random_state=42 for reproducibility. Permutation

tests use seeded `numpy.random`.

Blinding: N/A (computational analysis only; no investigator-blinded experiments).

4.5 ChIP-seq / Flow cytometry / Magnetic resonance / etc.

N/A — this study is computational re-analysis of public datasets only.

5. Expanded reference list (40+ entries, NComms-friendly)

The current draft includes 22 references. Expansion to 40+ for NComms-friendly literature coverage. Bold = newly added (vs Cell Rep Med draft). Items marked **[verify before submission]** need PubMed/DOI confirmation.

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